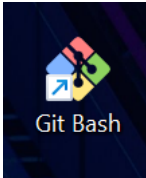


1. Git-HOL

Step 1: Setup your machine with Git Configuration

I have Git already installed and setup in my laptop



```
C:\Users\msaad>git --version
git version 2.49.0.windows.1

C:\Users\msaad>
```

```
MINGW64:/c/Users/msaad

msaad@Saad MINGW64 ~
$ git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=openssl
http.sslcainfo=C:/Program Files/Git/mingw64/etc/ssl/certs/ca-bundle.crt
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=main
core.editor="C:\Users\msaad\AppData\Local\Programs\Microsoft VS Code\bin\code" -
-wait
user.name=saadm4n
user.email=saadm4n@gmail.com
```

In Command prompt,

```
D:\>mkdir GitDemo

D:\>dir
Volume in drive D has no label.
Volume Serial Number is 7C24-420D
```

```
D:\>cd GitDemo

D:\GitDemo>git init
Initialized empty Git repository in D:/GitDemo/.git/

D:\GitDemo>echo "Welcome to Git hands-on lab" > welcome.txt

D:\GitDemo>ls
welcome.txt

D:\GitDemo>git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    welcome.txt

nothing added to commit but untracked files present (use "git add" to track)

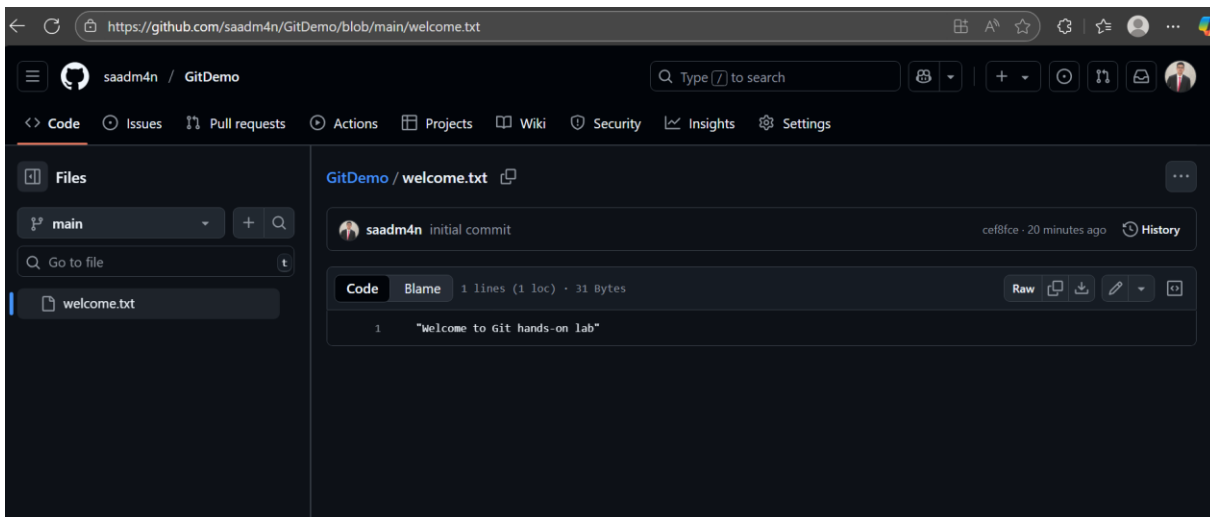
D:\GitDemo>git add .

D:\GitDemo>git commit -m "initial commit"
[main (root-commit) cef8fce] initial commit
1 file changed, 1 insertion(+)
create mode 100644 welcome.txt
```

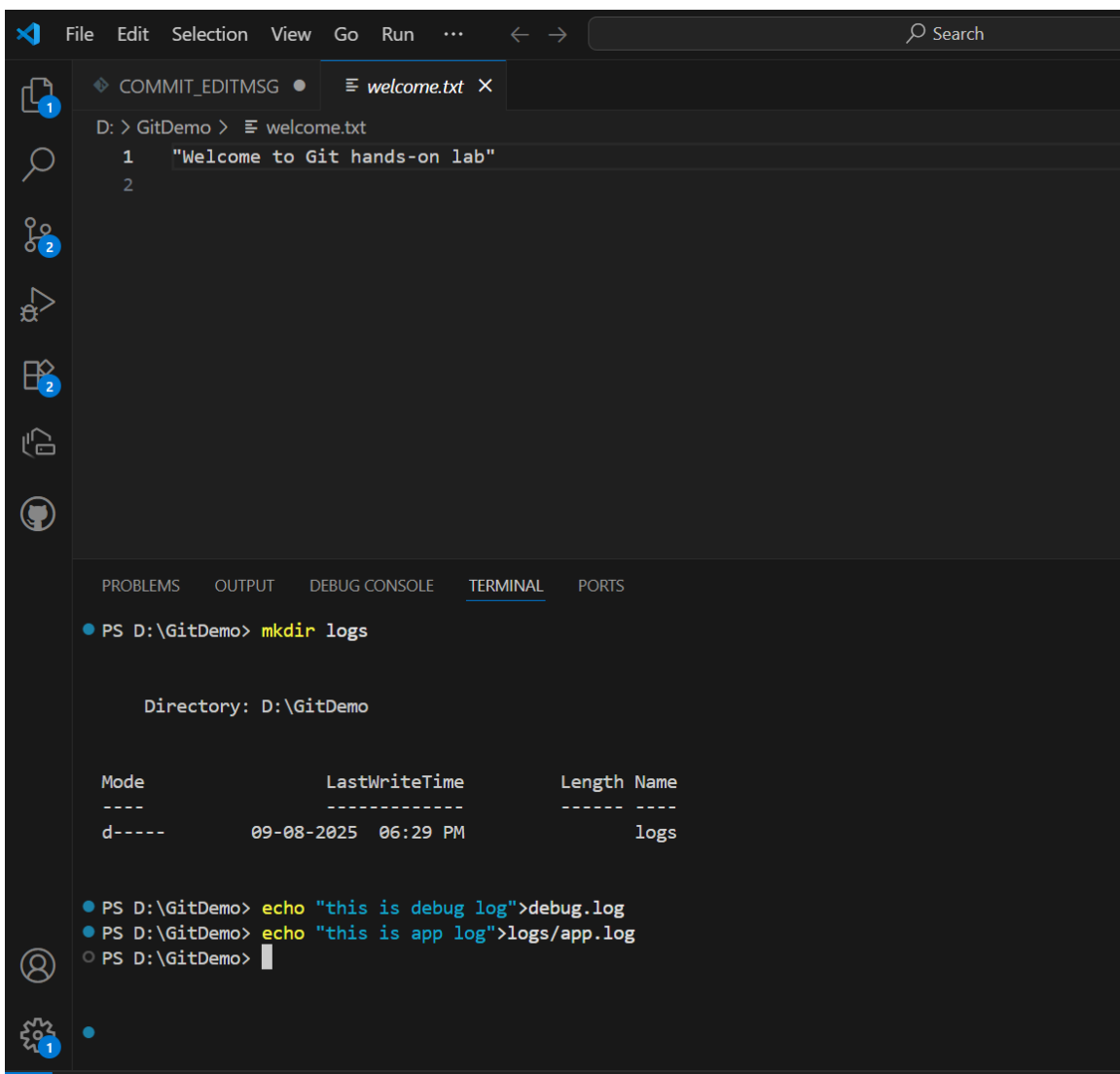
In VSCode

```
PS D:\GitDemo> git remote add origin https://github.com/saadm4n/GitDemo.git
● >>
● PS D:\GitDemo> git push -u origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 239 bytes | 239.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/saadm4n/GitDemo.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
○ PS D:\GitDemo> █
```

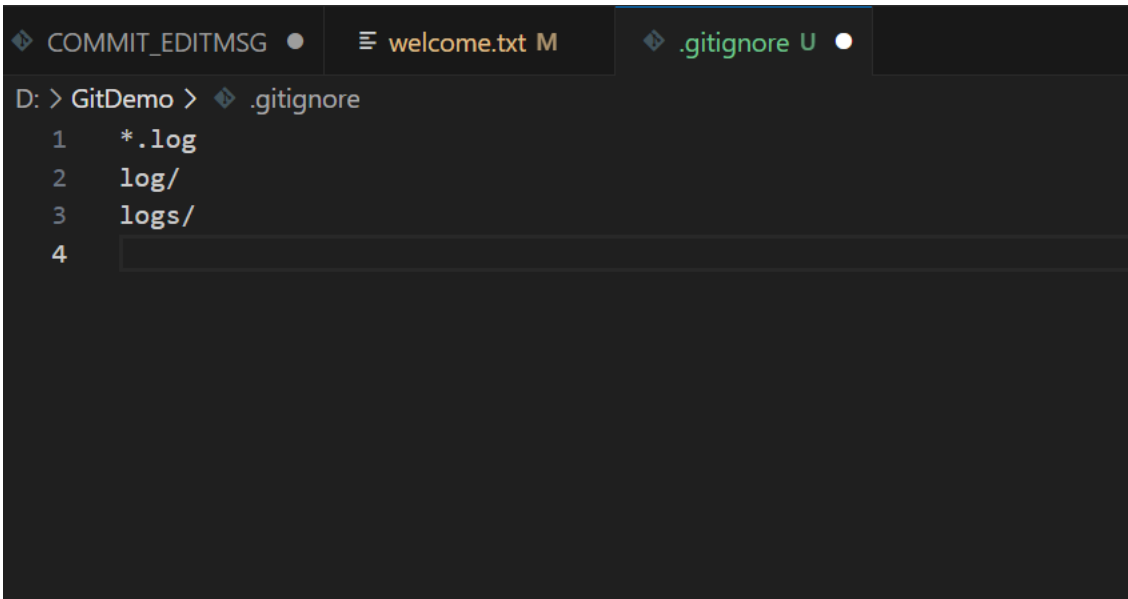
Output:



2. Git-HOL



Create a file “.gitignore”



The screenshot shows a code editor with three tabs: 'COMMIT_EDITMSG', 'welcome.txt M', and '.gitignore U'. The active tab is '.gitignore U', which contains the following text:

```
D: > GitDemo > .gitignore
1 *.log
2 log/
3 logs/
4
```

```
● PS D:\GitDemo> git add .gitignore
● PS D:\GitDemo> git commit -m "Add gitignore to ignore log files and folders"
[main 65bf411] Add gitignore to ignore log files and folders
1 file changed, 3 insertions(+)
create mode 100644 .gitignore
○ PS D:\GitDemo>
```

```
● PS D:\GitDemo> git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
(use "git push" to publish your local commits)

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   welcome.txt

no changes added to commit (use "git add" and/or "git commit -a")
○ PS D:\GitDemo>
```

Now, we can see that git is ignoring the log files

```
● PS D:\GitDemo> git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 20 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 305 bytes | 305.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/saadm4n/GitDemo.git
 cef8fce..65bf411  main -> main
○ PS D:\GitDemo>
```

3. Git-HOL

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

● PS D:\GitDemo> git checkout main
M      welcome.txt
Already on 'main'
Your branch is up to date with 'origin/main'.
⊗ PS D:\GitDemo> git branch GitNewBranch
fatal: a branch named 'GitNewBranch' already exists
● PS D:\GitDemo> git checkout
M      welcome.txt
Your branch is up to date with 'origin/main'.
● PS D:\GitDemo> git branch
  GitNewBranch
* main
● PS D:\GitDemo> git checkout GitNewBranch
M      welcome.txt
Switched to branch 'GitNewBranch'
```

```

PS D:\GitDemo> git branch
  GitNewBranch
* main
PS D:\GitDemo> git checkout GitNewBranch
M       welcome.txt
Switched to branch 'GitNewBranch'
PS D:\GitDemo> git branch
* GitNewBranch
  main
PS D:\GitDemo> echo "this is the content for the new branch">branchfile.txt
PS D:\GitDemo> git add .
PS D:\GitDemo> git commit -m "added file branchfile.txt to GitNewBranch"
[GitNewBranch f708eba] added file branchfile.txt to GitNewBranch
 2 files changed, 1 insertion(+)
 create mode 100644 branchfile.txt

```

```

PS D:\GitDemo> git status
On branch GitNewBranch
nothing to commit, working tree clean
PS D:\GitDemo> git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

```

```

PS D:\GitDemo> git diff GitNewBranch
diff --git a/branchfile.txt b/branchfile.txt
deleted file mode 100644
index 3b6eb4c..0000000
Binary files a/branchfile.txt and /dev/null differ
diff --git a/welcome.txt b/welcome.txt
index f47fea1..0918b2a 100644
--- a/welcome.txt
+++ b/welcome.txt
@@ -1,2 +1 @@
 "Welcome to Git hands-on lab"
-
PS D:\GitDemo> git merge GitNewBranch
Updating 65bf411..f708eba
Fast-forward
 branchfile.txt | Bin 0 -> 82 bytes
 welcome.txt    | 1 +
 2 files changed, 1 insertion(+)

```

```

PS D:\GitDemo> git branch -d GitNewBranch
Deleted branch GitNewBranch (was f708eba).
PS D:\GitDemo> git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean

```

4. Git-HOL

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

● PS D:\GitDemo> git branch
* main
● PS D:\GitDemo> git checkout main
Already on 'main'
Your branch is ahead of 'origin/main' by 1 commit.
(use "git push" to publish your local commits)
● PS D:\GitDemo> git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
(use "git push" to publish your local commits)

nothing to commit, working tree clean
● PS D:\GitDemo> git checkout -b GitWork
Switched to a new branch 'GitWork'
● PS D:\GitDemo> echo "<msg>Hello frm branch</msg>">hello.xml
● PS D:\GitDemo> git add hello.xml
● PS D:\GitDemo> git commit -m "added hello.xml"
[GitWork 6a248f8] added hello.xml
```

```

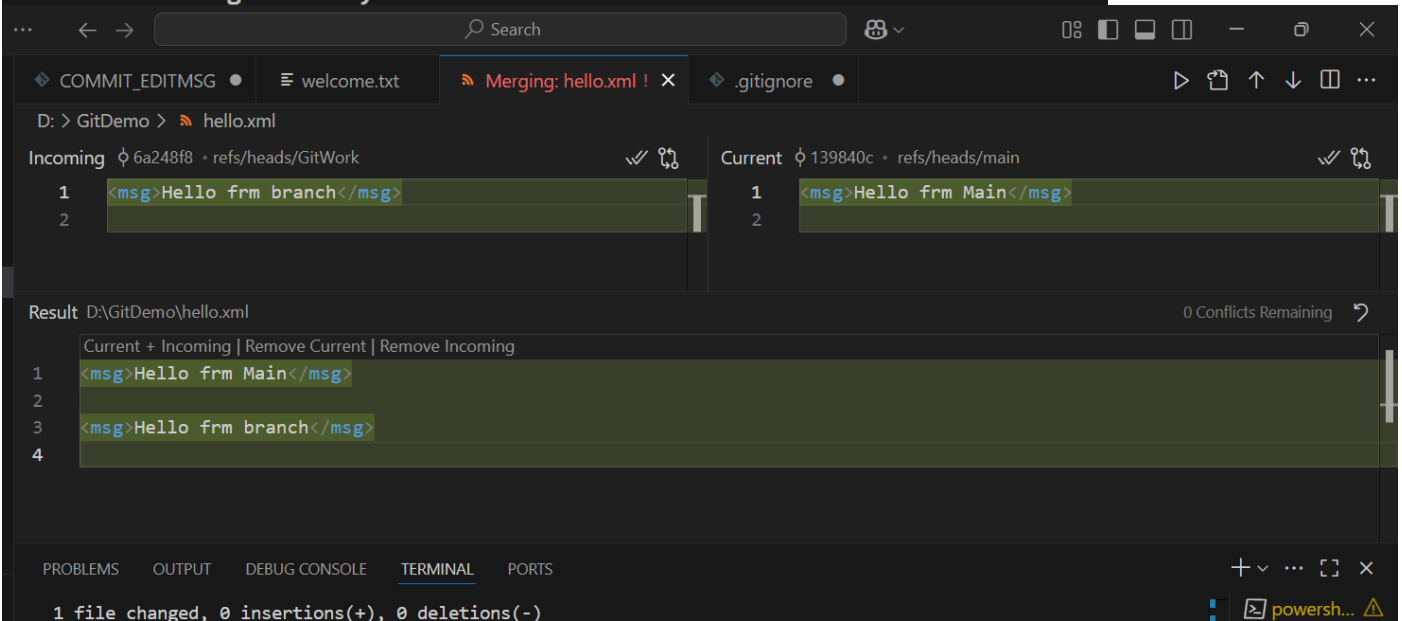
PS D:\GitDemo> git commit -m "added hello.xml"
[GitWork 6a248f8] added hello.xml
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 hello.xml
PS D:\GitDemo> git checkout main
Switched to branch 'main'
Your branch is ahead of 'origin/main' by 1 commit.
(use "git push" to publish your local commits)
PS D:\GitDemo> echo "<msg>Hello frm Main</msg>">hello.xml
PS D:\GitDemo> git add hello.xml
PS D:\GitDemo> git commit -m "added hello.xml"
[main 139840c] added hello.xml
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 hello.xml

```

```

PS D:\GitDemo> git merge GitWork
warning: Cannot merge binary files: hello.xml (HEAD vs. GitWork)
Auto-merging hello.xml
CONFLICT (add/add): Merge conflict in hello.xml
Automatic merge failed; fix conflicts and then commit the result.

```



I have resolved the conflict by choosing accept the incoming change


```
● PS D:\GitDemo> git add hello.xml
● PS D:\GitDemo> git commit -m "resolved conflict in hello.xml"
[main 3b8c48f] resolved conflict in hello.xml
● PS D:\GitDemo> git branch -d GitWork
Deleted branch GitWork (was 6a248f8).
● PS D:\GitDemo> git push origin main
Enumerating objects: 15, done.
Counting objects: 100% (15/15), done.
Delta compression using up to 20 threads
Compressing objects: 100% (12/12), done.
Writing objects: 100% (13/13), 1.11 KiB | 570.00 KiB/s, done.
Total 13 (delta 5), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (5/5), done.
To https://github.com/saadm4n/GitDemo.git
65bf411..3b8c48f main -> main
● PS D:\GitDemo>
```

5. Git-HOL

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

● PS D:\GitDemo> git checkout main
Already on 'main'
Your branch is up to date with 'origin/main'.
● PS D:\GitDemo> git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
● PS D:\GitDemo> git branch -a
* main
  remotes/origin/main
● PS D:\GitDemo> git pull origin main
From https://github.com/saadm4n/GitDemo
 * branch          main          -> FETCH_HEAD
Already up to date.
● PS D:\GitDemo> git push origin main
● Everything up-to-date
○ PS D:\GitDemo> 
```

Thank You
